

Setup using the Arduino IDE first as we will then be able to use the avrdude library installed for the Microchip Studio setup after.

1 Arduino Setup

Open your Arduino IDE desktop shortcut.

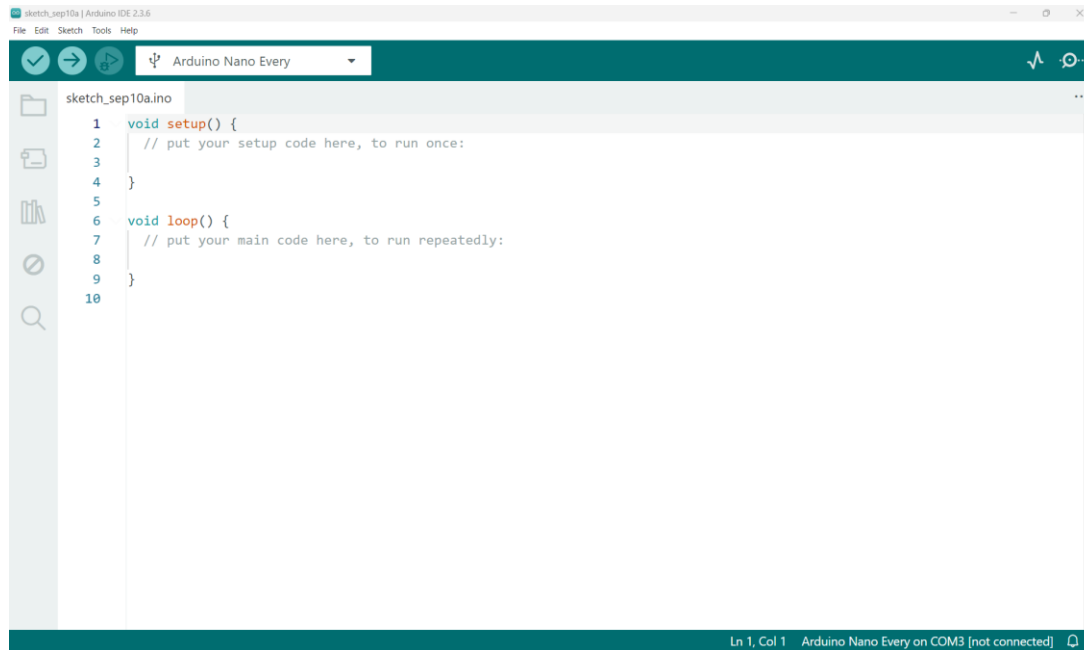


Figure 1: Arduino IDE Workspace

Plug in your data cable into the Nano Every device attached to Uno/UL OER Shield. There should be a constant green light on you device.

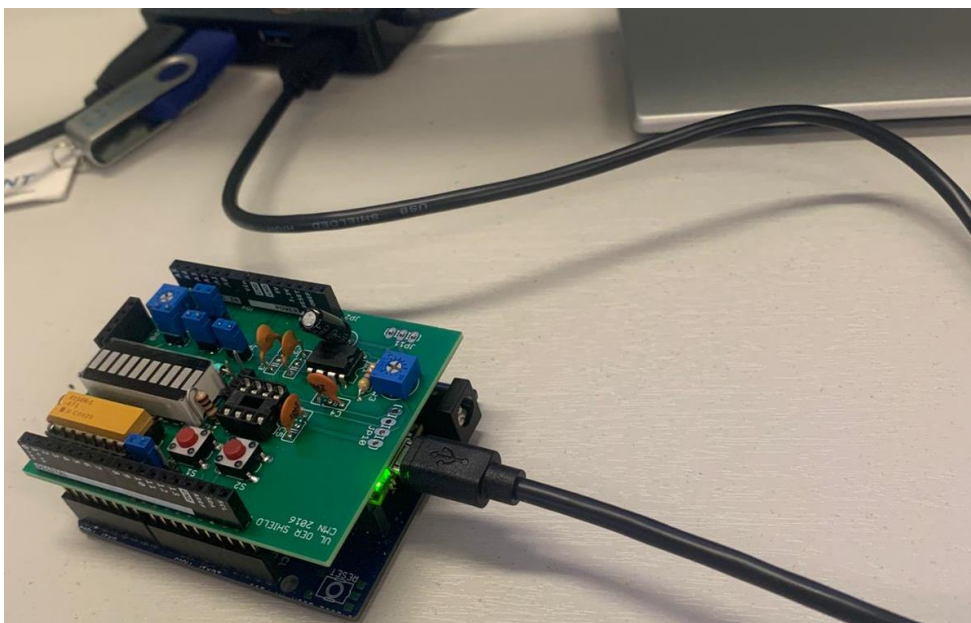


Figure 2: Device powered on (green light)

Click on connected Devices and there should be an Arduino Nano Every Option. Click on this to connect. The text should now be in bold font if you are fully connected.

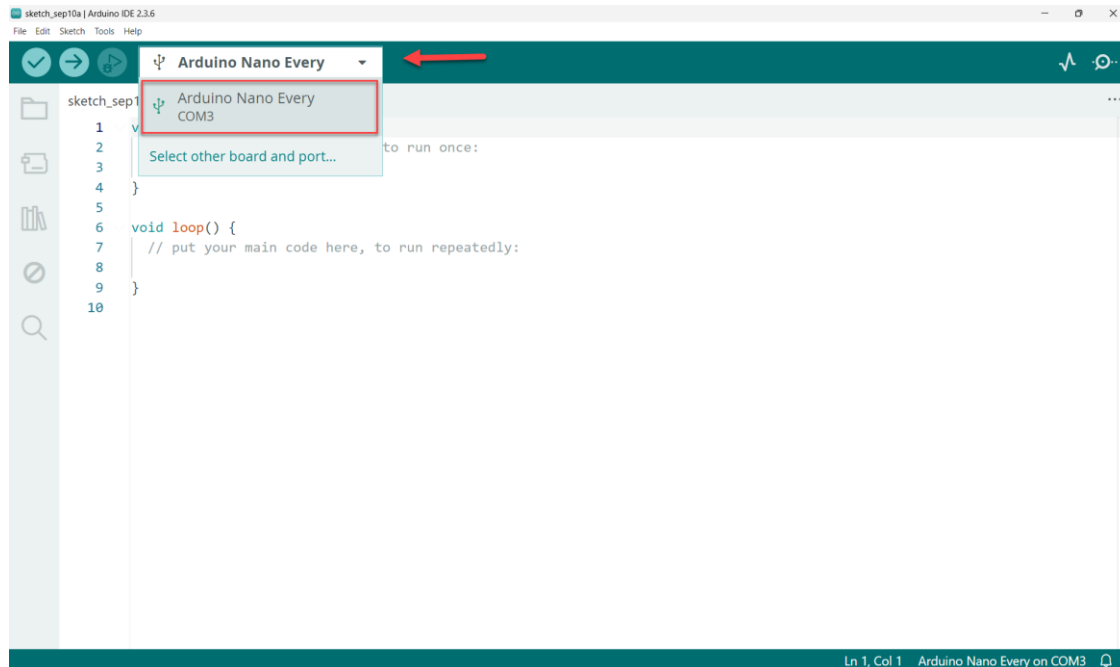


Figure 3: Connect to Arduino Nano Every

Go to the boards manager tab and install these two packages below by searching the term **Arduino AVR Boards**. Accept all permissions to install the required folder for this device. Avrdude and it's required libraries should now be installed on your machine.

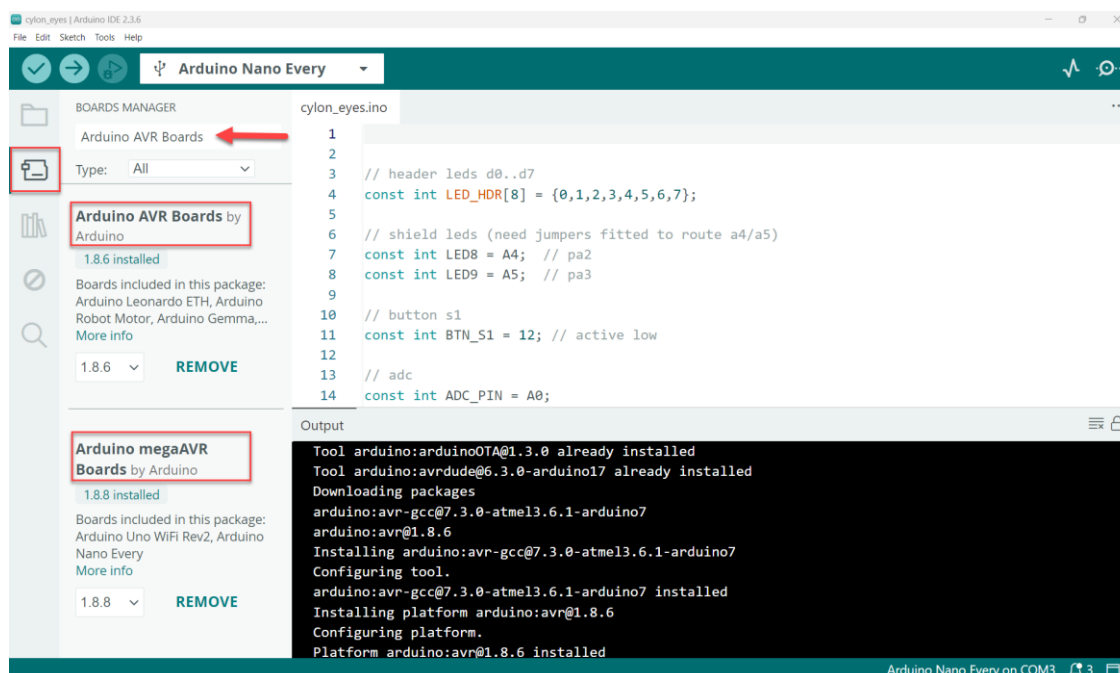


Figure 4: Install AVR board libraries for the connected board

Paste this code into your Arduino IDE and execute/upload it to ensure that your device has been setup properly.

```
/*
 * Lab 1 - Turn ON all shield LEDs Test (Arduino IDE C).
 * LEDs are ACTIVE-HIGH (write 1 = ON).
 * Author: Luke Griffin
 */

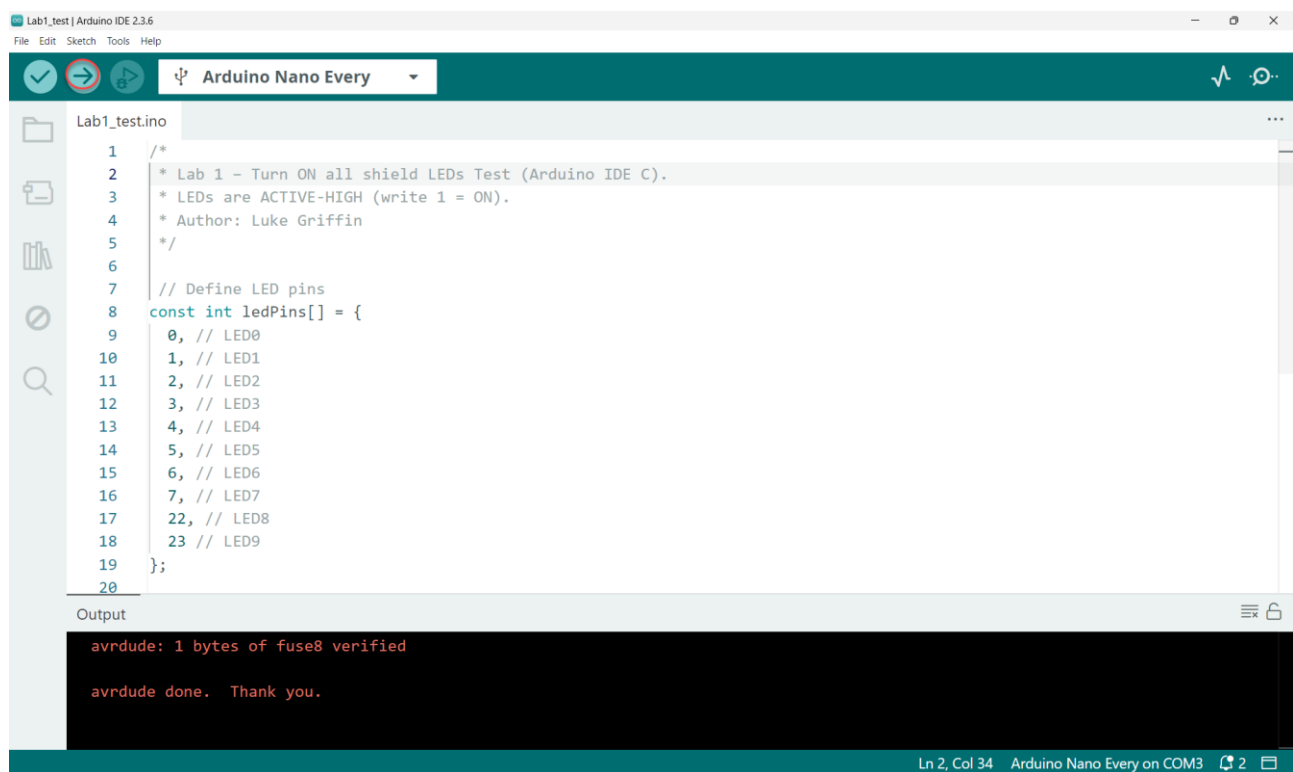
// Define LED pins
const int ledPins[] = {
  0, // LED0
  1, // LED1
  2, // LED2
  3, // LED3
  4, // LED4
  5, // LED5
  6, // LED6
  7, // LED7
  22, // LED8
  23 // LED9
};

const int numLeds = sizeof(ledPins) / sizeof(ledPins[0]);

void setup() {
  // Set all LED pins as OUTPUT
  for (int i = 0; i < numLeds; i++) {
    pinMode(ledPins[i], OUTPUT);
  }

  // Turn all LEDs ON
  for (int i = 0; i < numLeds; i++) {
    digitalWrite(ledPins[i], HIGH); // active-high LEDs
  }
}

void loop() {
  // Do nothing, LEDs stay ON
}
```



```
1  /*
2  * Lab 1 - Turn ON all shield LEDs Test (Arduino IDE C).
3  * LEDs are ACTIVE-HIGH (write 1 = ON).
4  * Author: Luke Griffin
5  */
6
7  // Define LED pins
8  const int ledPins[] = {
9      0, // LED0
10     1, // LED1
11     2, // LED2
12     3, // LED3
13     4, // LED4
14     5, // LED5
15     6, // LED6
16     7, // LED7
17     22, // LED8
18     23 // LED9
19 };
20
```

Output

```
avrdude: 1 bytes of fuse8 verified
avrdude done.  Thank you.
```

Ln 2, Col 34 Arduino Nano Every on COM3 2

Figure 5: Upload code

LED's should all be in the ON state on you shield.

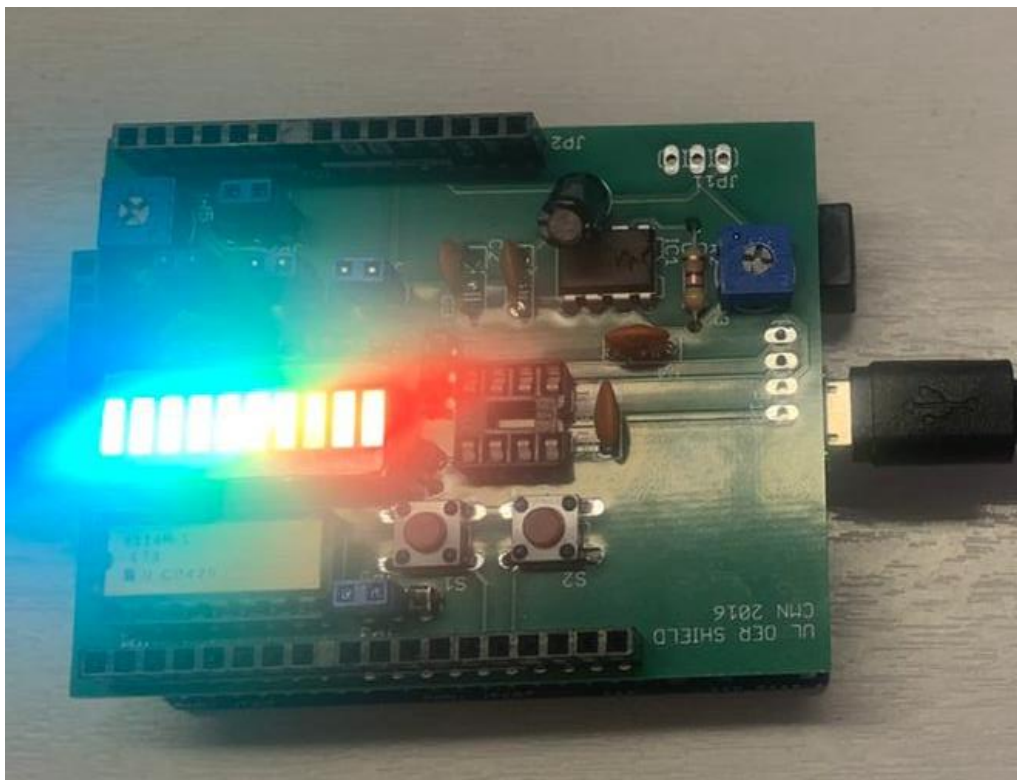


Figure 6: Result of test script execution